

International Airport Management System

Laboratory Work 1: ERD Diagram

Course: Databases

1. Title & Executive Summary

- **Project Title:** International Airport Database System
- **Purpose:** To design a relational database structure for managing airport operational processes including flight schedules, passenger profiles, booking records, boarding pass issuance, baggage tracking, and security screenings.
- **Submission Format:** PDF document containing the System ER-Diagram and structural textual description.

2. List of Identified Entities

The database system comprises 10 core entities:

- airport
- airlines
- passengers
- flights
- bookings
- booking_changes
- boarding_passes
- baggage
- baggage_checks
- security_checks

3. Entity Attributes and Data Types

- **airports:** airport_id (SERIAL, PK), airport_name (VARCHAR(100)), country (VARCHAR(100)), state (VARCHAR(100)), city (VARCHAR(100)), created_at (TIMESTAMP), updated_at (TIMESTAMP).
- **airlines:** airline_id (SERIAL, PK), airline_code (VARCHAR(10), UNIQUE), name (VARCHAR(100)), country (VARCHAR(100)), created_at (TIMESTAMP), updated_at (TIMESTAMP).
- **passengers:** passenger_id (SERIAL, PK), first_name (VARCHAR(100)), last_name (VARCHAR(100)), gender (VARCHAR(20)), date_of_birth (DATE), country_of_citizenship

(VARCHAR(100)), country_of_residence (VARCHAR(100)), passport_number (VARCHAR(50), UNIQUE), created_at (TIMESTAMP), updated_at (TIMESTAMP).

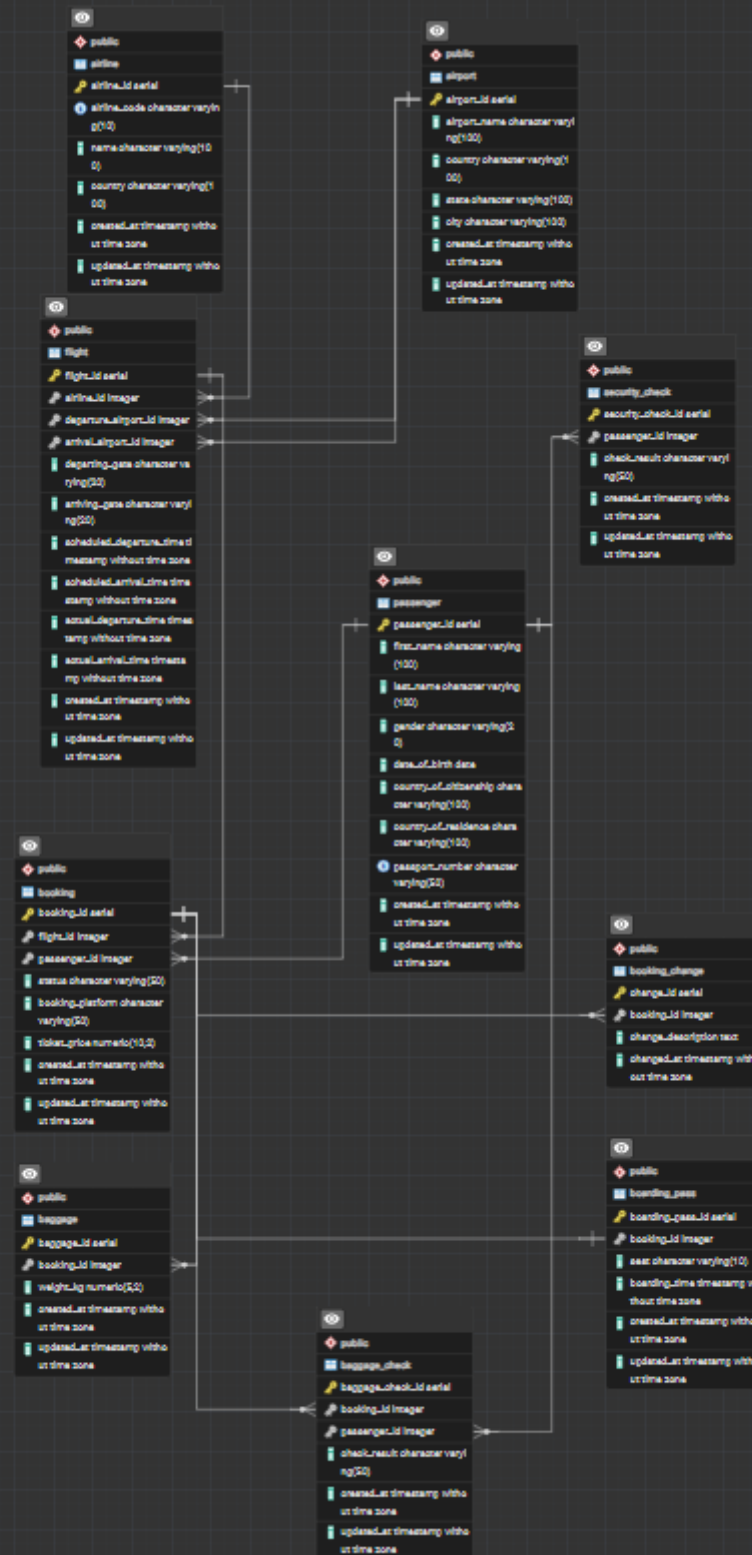
- **flights:** flight_id (SERIAL, PK), airline_id (INTEGER, FK), departure_airport_id (INTEGER, FK), arrival_airport_id (INTEGER, FK), departing_gate (VARCHAR(20)), arriving_gate (VARCHAR(20)), scheduled_departure_time (TIMESTAMP), scheduled_arrival_time (TIMESTAMP), actual_departure_time (TIMESTAMP), actual_arrival_time (TIMESTAMP), created_at (TIMESTAMP), updated_at (TIMESTAMP).
- **bookings:** booking_id (SERIAL, PK), flight_id (INTEGER, FK), passenger_id (INTEGER, FK), status (VARCHAR(50)), booking_platform (VARCHAR(50)), ticket_price (NUMERIC(10,2)), created_at (TIMESTAMP), updated_at (TIMESTAMP).
- **booking_changes:** change_id (SERIAL, PK), booking_id (INTEGER, FK), change_description (TEXT), changed_at (TIMESTAMP).
- **boarding_passes:** boarding_pass_id (SERIAL, PK), booking_id (INTEGER, FK, UNIQUE), seat (VARCHAR(10)), boarding_time (TIMESTAMP), created_at (TIMESTAMP), updated_at (TIMESTAMP).
- **baggage:** baggage_id (SERIAL, PK), booking_id (INTEGER, FK), weight_kg (NUMERIC(5,2)), created_at (TIMESTAMP), updated_at (TIMESTAMP).
- **baggage_checks:** baggage_check_id (SERIAL, PK), booking_id (INTEGER, FK), passenger_id (INTEGER, FK), check_result (VARCHAR(50)), created_at (TIMESTAMP), updated_at (TIMESTAMP).
- **security_checks:** security_check_id (SERIAL, PK), passenger_id (INTEGER, FK), check_result (VARCHAR(50)), created_at (TIMESTAMP), updated_at (TIMESTAMP).

4. Relationships and Cardinalities

- **airlines → flights (1: N):** One airline operates multiple scheduled flights.
- **airports → flights (1: N):** One airport serves as departure or arrival location for multiple flights.
- **flights → bookings (1: N):** One flight receives multiple passenger bookings.
- **passengers → bookings (1: N):** One passenger can make multiple flight reservations.
- **bookings → booking_changes (1: N):** One booking can have multiple change history logs.
- **bookings → boarding_passes (1: 1):** Each valid booking generates exactly one boarding pass.
- **bookings → baggage (1: N):** One booking can include zero or more registered baggage items.

- **bookings & passengers → baggage_checks ($N: 1$):** Inspection records reference the corresponding booking and passenger.
- **passengers → security_checks ($1: N$):** A passenger undergoes security checks prior to departure.

5. Entity-Relationship Diagram (ERD)



public
airline
airline_id serial
airline_code character varying(10)
name character varying(100)
country character varying(100)
created_at timestamp without time zone
updated_at timestamp without time zone

public
airport
airport_id serial
airport_name character varying(100)
country character varying(100)
state character varying(100)
city character varying(100)
created_at timestamp without time zone
updated_at timestamp without time zone

public
flight
flight_id serial
airline_id integer
departure_airport_id integer
arrival_airport_id integer
departing_gate character varying(20)
arriving_gate character varying(20)
scheduled_departure_time timestamp without time zone
scheduled_arrival_time timestamp without time zone
actual_departure_time timestamp without time zone
actual_arrival_time timestamp without time zone
created_at timestamp without time zone
updated_at timestamp without time zone

public
passenger
passenger_id serial
first_name character varying(100)
last_name character varying(100)
gender character varying(20)
date_of_birth date
country_of_citizenship character varying(100)
country_of_residence character varying(100)
passport_number character varying(50)
created_at timestamp without time zone
updated_at timestamp without time zone

public
security_check
security_check_id serial
passenger_id integer
check_result character varying(50)
created_at timestamp without time zone
updated_at timestamp without time zone

public
booking
booking_id serial
flight_id integer
passenger_id integer
status character varying(50)
booking_platform character varying(50)
ticket_price numeric(10,2)
created_at timestamp without time zone
updated_at timestamp without time zone

public
booking_change
change_id serial
booking_id integer
change_description text
changed_at timestamp without time zone

public
baggage
baggage_id serial
booking_id integer
weight_kg numeric(5,2)
created_at timestamp without time zone
updated_at timestamp without time zone

public
baggage_check
baggage_check_id serial
booking_id integer
passenger_id integer
check_result character varying(50)
created_at timestamp without time zone
updated_at timestamp without time zone

public
boarding_pass
boarding_pass_id serial
booking_id integer
seat character varying(10)
boarding_time timestamp without time zone
created_at timestamp without time zone
updated_at timestamp without time zone

6. Diagram Key & Legend

- **PK (Primary Key):** Unique identification key for each table entry.
- **FK (Foreign Key):** Referencing key establishing relational integrity with another table's primary key.
- **1: N (One-to-Many):** One record in table A relates to multiple records in table B.
- **1: 1 (One-to-One):** Each record in table A corresponds to a single unique record in table B.

7. Solution Summary & Description

The database architecture provides a fully normalized structure (3NF) designed to support complex airport workflows. Central operations revolve around the flights and bookings tables, linking operational logistics (airlines and airports) with customer interactions (passengers, baggage, boarding passes, and security checks). Audit trails for flight modifications are preserved via the dedicated booking_changes entity.